WORKING PAPER / PRELIMINARY DEVELOPMENT EVIDENCE

Sequential patch acquisition for lunar camera localization

A small measured lead, a clean-image evaluation gap, and a precise contribution hypothesis.

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Abstract

We study whether sequential native-image measurements can recover lunar camera poses that a fixed measurement set misses. A classical shared-camera correlation pipeline with a reference-height-aware selection proxy recovers 8/16 raw poses, compared with 7/16 for uniform and geometry policies at the same 32-extra-search budget. The isolated 6.25-point gain has a descriptive terrain interval including zero. Active selection is about 8.33 times slower than Direct. The evidence reuses four supplied-light acquisition conditions on three terrain components; it is not a changing-light generalization result.

A separate unknown-light catalogue/camera initializer passes generated controls, but its measured cost forecast prevents lunar inference within the frozen cap. Historical modern comparisons use a different, default-dither image pipeline and cannot establish a clean ranking for the new method. We therefore present an audited diagnostic study and an unproven mechanism hypothesis, rather than a superior learned matcher.

1. Research question

At a fixed native patch-search budget, can measurements selected to distinguish globally consistent camera and illumination hypotheses improve correct absolute camera localization despite reference-map uncertainty? The eventual compact learned system remains constrained to fewer than 100,000 total parameters; the presently measured acquisition component adds no learned parameters.

2. What the present evidence contributes

The completed result separates extra measurement support from selection policy, retains initialization failures, compares all fresh methods with a common pose solver and independently replays every saved score. The code also exposes a remaining primary-track gap: removing supplied matching light requires a larger camera/light search whose lunar recovery has not been measured.

No flight-readiness, orbital accuracy, calibrated 1% risk, new-terrain generalization or novelty claim follows. The candidate contribution concerns a specific physical measurement policy; broad active vision, learned refinement and compact lunar reconstruction have relevant precedents.

Keywords: lunar terrain-relative navigation; active measurement; camera pose; illumination; reproducibility.

METHOD

Select measurements, then update

The measured policy remains frozen; the unknown-light extension is a separate untested lunar configuration.

3. Native correlation and camera fitting

Inputs are native 512-pixel reference/query images, reference depth, known reference camera-to-world pose and both camera intrinsics. Reference geometry lifts image anchors into 3D. The query true pose, depth and Sun are scoring-only. The measured pipeline begins with 36 native correlation fields and a shared camera fit to their ambiguity-preserving image evidence.

4. Sequential acquisition

Form 256 spatial reference candidates with eligible geometry, excluding locations too close to acquired anchors. Around the current camera, construct seven bounded hypotheses along weak projection-Jacobian directions. Cheap first-order reference previews predict observation changes under these hypotheses. For each anchor, repeat prediction under 0,+/-5,+/-10,+/-20 m reference-height changes along local up expressed in the reference frame.

The active score favors separable prediction envelopes. Unknown evidence gives no supporting discrimination and retains its denominator weight. This is a conservative proxy; it is not calibrated mutual information or a probability of correctness. Select a candidate within the shared spatial slot, acquire its actual query correlation field, update the camera and repeat. Unselected candidates receive no expensive query-field measurement. Failed attempts consume searches.

5. Equal-search controls and final pose

Uniform, reference-appearance, geometry-only and active policies share candidate eligibility, spatial slots, extra-search limits, extra-template construction, camera updates and final estimation. Only comparisons among these policies isolate selection. The no-extra comparison also changes the measurement and template path, so its full gain is not attributable to the active score.

Use camera-conditioned native-grid modes, local peak refinement and frozen appearance/ambiguity/reverse checks. The final AP3P/refit requires six matches and a three-native-pixel reprojection threshold. The soft camera proposal never replaces a failed final solve. Public uncalibrated poses remain null; raw proposals are explicit research outputs.

6. Unknown-light extension

A separate initializer ranks up to 24 reference illuminations across 2,048 reference-derived camera hypotheses using full 3D z-buffer projections and common photometric support. It refines at most eight seeds and chooses one shared catalogue light from image evidence. This is discrete illumination selection, not continuous Sun estimation or independent solar-visibility validation. The retained catalogue light conditions the same native acquisition controls.

Keep the image pipelines separate

A changed input encoding or shared terrain can change what a comparison establishes.

7. Completed acquisition cohort

The measured v137 cohort contains 16 positives and 16 original nonoverlaps. Four exposed acquisition conditions cross two independent reference rendering-noise draws with two query draws. There is one faint, one bright and two nearly uniform conditions; the two uniform conditions share one terrain component. The effective geography comprises only three components. Matching reference illumination is supplied; historical light-index identifiers do not make this an unknown-light experiment.

Evidence	Input / scope	What may be concluded
Historical modern comparison	48 positives;8 terrains; default-dither RGB8	Recorded performance on that encoding only
Active acquisition pilot	16 positives;3 terrains; supplied-light dither-free EXR	A small within-pilot policy difference
Clean unknown-light roster	4 positives +4 nonoverlaps; RGB16;104 images	Inputs prepared; no lunar predictions
Full-budget generated controls	Flat/layered pattern fixtures;24 lights;2,048 cameras	Software/cost evidence, not physical sunlight robustness

8. Why default dithering matters

Earlier audits found detector-coordinate image structure from Blender default dithering in dark renders. A featureless emission-plane control produced 2,025 same-coordinate feature matches with default dithering and zero after disabling it. A targeted four-positive paired clean-image study retained the bright success, removed all three wrong low-light poses and recovered none of the three dark positives. This identifies an encoding failure mechanism; it does not determine the full clean modern ranking.

The old 48-positive modern benchmark remains a reproducibility record: Direct 30/48, LoMa-R15/48 and RoMa v2 15/48. The new active pipeline has not run that roster. Only 3/16 reference cameras in the larger roster currently have complete clean 24-light catalogues. Thirteen need clean acquisition. Neither those historical counts nor heterogeneous paper percentages can rank the current active method.

9. Metrics and protected data

Correct raw pose requires rotation error<=2 degrees and absolute translation error<=1% of median Euclidean reference range. Strict pose uses 0.5 degrees and 0.1%. Retain all eligible positives, including no-initializer/no-pose cases. Native correspondence thresholds are 1/2/4/8/12 original pixels. Four calibration and twelve final terrain groups remain closed; there is no independent orbital pose reference or human landmark review.

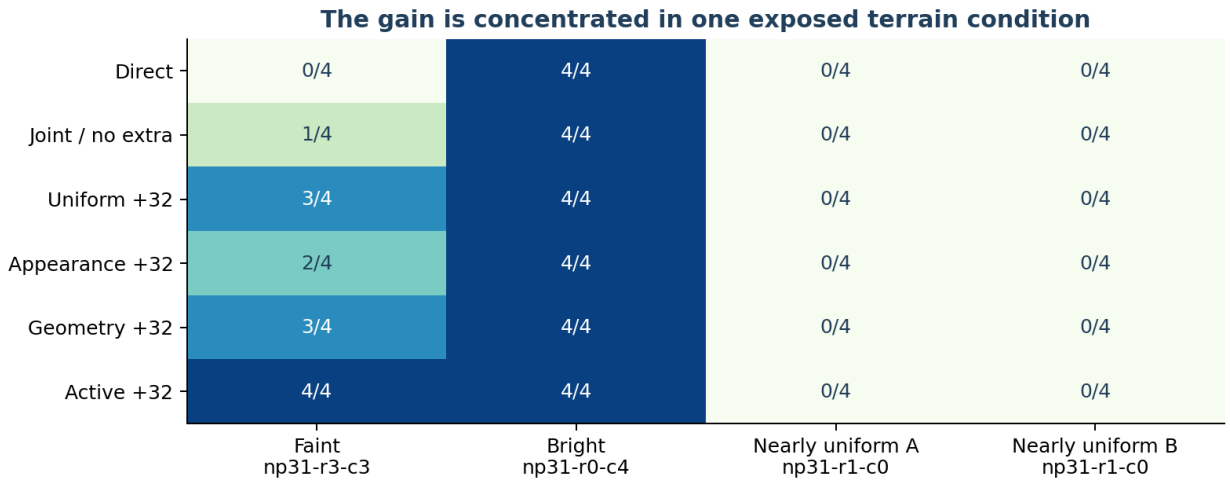
MEASURED ACQUISITION RESULT

One extra pose over equal-search controls

The prospective early signal passes; the original accepted-pose research targets do not.

Method	Correct /16	Wrong	Strict /16	Seconds
Direct	4	0	4	1.36
Joint / no extra	5	1	4	7.52
Uniform +32	7	1	4	11.11
Appearance +32	6	4	4	10.55
Geometry +32	7	1	5	11.60
Active +32	8	0	4	11.31

Seconds are positive-pair online medians including fresh initialization and decoding, with reference-only caches permitted. All six methods return zero raw poses on 16 nonoverlaps. Worker memory includes shared caches and cannot rank individual methods.



10. Effect size and uncertainty

Raw gains are 25/18.75/6.25 points over Direct/no-extra/best policy; only the last isolates selection. The 27 three-component bootstrap draws give descriptive intervals[0,78.33] points over Direct and[0,19.58] over uniform/geometry. Both include zero; population evidence remains limited.

Active recovers all four faint noise pairings versus three for uniform/geometry. Bright cases already succeed; all eight nearly uniform positives fail. Geometry leads strict recovery 5/16 versus active 4/16. Active has 243 two-pixel matches versus uniform 244; more precise matches alone do not explain the pose result.

Negative resampling weights both terrain endpoints; nine of 27 draws have no edges. Zero observed errors and an all-zero bootstrap cannot establish a 1% deployment risk.

11. Verification

All 832 saved method/case cells independently replay. The fresh initializer and Direct match saved controls exactly; all 20 historical controls remain unchanged, and eight available no-extra rows repeat the prior pilot. These are reproducibility checks on consumed data, not independent terrain validation.

The primary experiment is still unrun

Successful generated controls establish wiring; their cost prevented the full lunar comparison.

12. Full-budget control result

Control	Complete pair time	Configuration
Flat geometry	46.06 s	All six methods;24 references;2,048 camera proposals
Layered geometry	51.88 s	Same configuration;32 extra searches per acquisition policy
Full 8-case forecast	652.51 s	1.5 x8 x worst pair +30 s margin

Every generated arm recovered its known camera within the frozen control tolerance. The fixtures contain arbitrary image patterns, not a physical sunlight simulation. The measured forecast exceeds even the full 480-second recovery cap before the pilot runtime. The controller stopped before lunar prediction, so this is not 0/4 or 0/8 lunar recovery. No smaller substitute was run.

RGB16 is decoded without RGB8 truncation and converted from Standard/sRGB to linear luminance. Geometry support is independent of brightness. Camera/light scores and query measurements are fresh; only reference projections/descriptors are cached. The shared illumination output remains a catalogue proposal, not a certified Sun estimate. The full 24-light grid also contains the tested query-light configurations; off-grid robustness is not established.

13. What the zero-pose work taught us

Earlier native measurements failed through weak candidate support and inadequate reliable correspondences. Prior reference-frame and measurement repairs restored 4/16, tying Direct. A later faint camera proposal came within 10.02 m but supplied only five reliable matches, below the six-match gate. Sequential extra acquisition can cross that support threshold on the faint condition. This is a measured pathway, not proof that the full prior 0/16 failure had a single cause.

14. Remaining mechanisms to distinguish

The present policy may help by measurement count, spatial arrangement, camera conditioning or local image ambiguity. Equal-budget controls are essential. A learned response model must add a gain beyond deterministic native refinement, not merely obtain more searches or more reference information. Height-envelope projections do not substitute for rerendering altered terrain/shadows or for validating solar visibility under missing map coverage.

No new training, calibration, final evaluation or sensor-transfer result is reported by this document update. Earlier software and data failures, frozen deadlines and resource charges remain preserved.

Compare the task before the number

Published scores describe different datasets, geometry access and error definitions.

Source	Published result / method	Relationship to this study
Lunar TRN [1]	Fine-tuned LoFTR unseen-region RA-1/RA-5: 70.6/85.9%	Match precision, not absolute pose success; lunar adaptation already exists
LoMa [4]	LoMa-G HardMatch 54.3; WxBS 73.4 mAA@10px	Not the LoMa-R profile in our CPU study
RoMa v2 [5]	MegaDepth 86.6; ScanNet 73.8 AUC@20deg	Relative-pose AUC; paper uses 800/1024px matching/refinement
RoMa-Omega [6]	Its Table 2: WxBS 72.9 vs 64.8 for RoMa v2	Within-paper comparator; A100 throughput is not our CPU latency
Lunar distillation [8]	Up to 7x smaller than 688M teacher; reduced student 94.9M	Compact lunar reconstruction is prior art; far above our 100k constraint

15. What is already established

LIMA derives illumination-aware evidence from co-registered references; the same paper separately adapts LoFTR to lunar imagery [1]. RoMa v2 supplies subpixel refinement and predicted covariance [5]. CoRe explicitly models coarse-assignment failures separately from fine noise and uses reliability-weighted refitting [9]. Active uncertainty-guided search [2] and learned featuremetric localization against 3D maps [3] are established ideas.

StereoLunar adapts MASt3R to lunar reconstruction/relative pose and releases MOONSt3R [7]. Recent lunar foundation-model distillation [8] uses SVD-based decoder initialization and feature transfer. Thus neither lunar fine-tuning, compression nor a reliability/refinement head alone is a sufficient new contribution.

16. Candidate distinction

The remaining hypothesis couples one physically consistent illumination explanation to camera-conditioned native measurements, then chooses observations robust to reference-map uncertainty. It is a candidate integration, not established novelty. A defensible claim needs a gain uniquely attributable to that coupling, beyond common reference information and equal-search deterministic controls. The current one-pose supplied-light lead is insufficient.

This is a targeted review, not an exhaustive novelty search. No published terrestrial matching score is converted into our lunar accepted-pose rate. No new untested method replaces a required baseline.

A falsifiable contribution, with the original targets

The project remains active; documentation quality does not satisfy the model-performance requirements.

17. Necessary discriminating comparisons

Keep the prepared clean eight-case study frozen: Direct; joint without extras; uniform, appearance, geometry and active with 32 extra searches. Estimate query light from images; do not add a true-query matching template. Retain all failures and original nonoverlaps. Seal every prediction before loading scoring geometry and independently replay all 48 cells. This tiny study diagnoses the primary-track change; it cannot establish generalization or 1% risk.

For the full contribution, evaluate all eight A/B/C on/off combinations and the pooled-response control with frozen checkpoints. Separate data, response representation, globally shared Sun and active measurement effects. Include an independent-Sun-per-correspondence control. Repeat the selected learned configuration with the three predefined seeds and preserve every outcome.

18. Original success targets

Requirement	Current evidence
>=70% correctly accepted poses	50% raw supplied-light recovery; no accepted-pose study
>=10-point accepted gain; positive terrain interval	6.25-point raw gain over strongest equal-search policy; interval includes 0
<=1% empirical accepted error/nonoverlap	Zero raw errors on a small dependent pilot; no independent calibration
Modern pose advantage or useful accuracy/speed tradeoff	No matched clean modern ranking; active slower than Direct
Fresh test, transfer and robustness	Untouched reserve preserved; unknown-light, full terrain and sensor claims open

19. Fair modern comparison

Complete the declared native comparisons against frozen/recalibrated CNN, deterministic refinement, SIFT, MatchAnything-EfficientLoFTR, LoMa-R, RoMa v2 and MOONSt3R where pretrained controls and resources qualify. Preserve author profiles and common-input profiles separately. Use identical original image content, reference geometry and pose estimation; map all endpoints to native pixels. Query depth/pose/Sun remain labels, except the separately declared supplied-Sun track.

Clean acquisition must precede a broader clean ranking. Record training-data overlap, both negative endpoints, reference preparation, latency and memory. Keep calibration/final groups closed until development supports a qualifying frozen candidate. An inconclusive or negative learned result remains a finding; it does not become a superior model through threshold adjustment.

Inspect the evidence, keep the claims bounded

Presentation generated from unchanged saved results and a dated primary-source review.

20. Reproduction

V137 release contains the 832-cell measured acquisition replay. V139 release contains two ordinary clean-input ZIPs and checksums; extract both into a fresh directory and run `python reports/solar-blind-v139/replay_cost.py`. This verifies 101 source/125 input hashes and cost arithmetic, not a saved generated-pose replay.

This paper builder imports no matcher, pose fitter, renderer or training code. The companion evidence file hashes the saved statistics, figures, inventory and primary-source notes. The protected 104-source/130-input continuation closure remains unchanged. The v137 brief and paper remain historical artifacts. No email or paper submission is automatic.

References

[1] Rothenberger et al. (2025). Illumination Invariant Image Matching for Lunar TRN. Primary source.

[2] Davison (2005). Active Search for Real-Time Vision. Primary source.

[3] Sarlin et al. (2021). Back to the Feature: Learning Robust Camera Localization from Pixels to Pose. Primary source.

[4] Nordstrom et al. (2026). LoMa: Local Feature Matching Revisited. Primary source.

[5] Edstedt et al. (2026). RoMa v2: Harder Better Faster Denser Feature Matching. Primary source.

[6] Nordstrom et al. (2026). RoMa-Omega: What Feed-Forward 3D Models Know About Image Matching. Primary source.

[7] Grethen et al. (2025). Adapting Stereo Vision From Objects To 3D Lunar Surface Reconstruction with the StereoLunar Dataset. Primary source.

[8] Grethen et al. (2026). Geometric Foundation Model Distillation for Efficient Lunar 3D Reconstruction. Primary source.

[9] Hoang et al. (2026). Semi-Dense Matching Uncertainty Is Not Just Local Confidence. Primary source.

20.1 Source and metric limitations

The LIMA numerical table was checked in the existing author copy (p.11); the JPL publication list verifies the DOI. The publisher endpoint was unavailable during this update. Other entries were checked against primary papers or author project pages on 6 November 2026. Website results and model profiles may differ from published paper configurations; the review records each specific source and metric.

Conclusion: additional native measurements repair some low-light support failures, while the active policy supplies only a small, inconclusive supplied-light gain. The next useful result must demonstrate the globally consistent illumination/measurement mechanism on clean data, with matched controls and preserved uncertainty limits.